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1.0 - Executive Summary

This objective of this project was to simulate the liquefaction of natural gas (LNG) processes through Aspen Plus and compare the annual operating costs of the Claude and Linde processes. A rigorous heat exchanger design was necessary to complete the project.

A detailed Process Flowsheet Diagram (PFD) can be found in Appendix 7.2. The process resulted in 298 kg/h of liquefied natural gas (LNG) being produced, with a loss of 151 kg/h to prevent Nitrogen from building up in the system. A detailed analysis of the compressors can be found in Appendix 7.3.1, and the heat and weight balance can be viewed in Appendix 7.3.2. TEMA sheets for the heat exchangers can be found in appendix 7.3.3.

The total capital cost of all equipment is \$16.4MM, and the yearly operating cost, or OPEX, is \$426M/yr. In total, the annual operating profit comes to \$1.2MM/yr. Detailed reports can be viewed in Appendix 7.3.4.

For the Linde process, the annual operating profit was found to be \$-358M/yr. Compared to the Claude process, the Linde process was found to be much less profitable. After running a sensitivity analysis, we found the optimal split fraction the expander to be 39%, however these results were unreliable due to the inaccurate sensitivity analysis graph, shown in Appendix 7.3.5.

2.0 – System and Procedure

The system is composed of two mixers, a three-stage compressor with intermediate cooling, one cooler, two heat exchangers, two splitters, one throttling valve, one expander and one separation tower. We assumed an annual operating year of 8000hrs/op-year.

The liquefaction of natural gas via the Claude process occurs from a fresh feed at 20°C and 1 bar pressure, composed mainly by methane (90.28% mole fraction) at a flowrate of 25kmol/h. The fresh feed is then combined with a recycling stream, known as the cold stream, coming from the first heat exchanger (E-102) and into the first mixer (MIX1). The resultant stream, S1, is then fed to the three-stage compressor system (K-100 A,B,C) with an inlet temperature of 35°C and 1 bar pressure and comes out of the last compressor as 142°C and 30 bar pressure.

The outlet stream, S2, is further cooled down by the pre-cooler (E-101) to 40°C prior to enter the heat exchanger. The outlet stream, S3, is then introduced to the first heat exchanger (E-102), where the hot stream is cooled down to 13°C by recycle stream S12. At the second heat exchanger (E-103), the resultant of the hot stream is cooled even further to -94°C by recycle stream S11. At this point, the outlet stream, S4, is sent to the first splitter (SPLIT1) in which 35% of the stream coming out of the cooler is sent to the expander (EX-100) that expands the split stream, S5, from 30 bar to 1 bar. This stream will be later combined with the vapor product of the separator (D-100) inside the second mixer (MIX2) and will be recycled back to the second heat exchanger.

The leftover stream, S6, from the first splitter is fed to the second heat exchanger where it will be cooled down with the recycling stream previously described. The temperature of S6 is reduced from 13°C to -94°C. The resultant stream, S7, from the second heat exchanger passes through the throttling valve where a reduction of pressure from 30 bar to 1 bar and results in a decrease of temperature from -94°C to -160°C. The stream, S8, is then fed to the separator where the liquified product, liquefied natural gas (LNG), is collected from the bottoms. The vapor product is sent to the second splitter (SPLIT2), set to a 10% split fraction, where a purge stream is collected in order to minimize the impurities in the system. The outlet from the second splitter, stream S14, is then combined in the second mixer (MIX2) with the outlet of the expander, completing the cycle. Please refer to Appendix 7.2 for the process flow diagram.

Based on the design specifications, the compressors and expander were set to 85% efficiency, and the discharge pressure of the third compressor was set to 30 bar. It was considered no pressure drop in the separator and an allowable 10psi pressure drop in the pre-cooler, inter-stage coolers and heat exchangers. The flowrate going to the expander was set as 35% of the flow of the hot stream entering the first heat exchanger. Fouling factor was also set as 0.002, which correspond to 0.001 on each side of the heat transfer area. The cooling water available is at 30°C. The minimum temperature approach was set to 10°C for the pre-cooler and inter-stage coolers.

To achieve this process, we set the hot inlet-cold outlet temperature difference to 2°C for E-102 and the cold stream outlet temperature to 3°C in E-103. We varied the split fraction in SPLIT2 until the mass balance was achieved in the system, which consisted of removing the inert

Nitrogen in the feed through the purge stream. Once the system converged, a TEMA sheet analysis was performed on both heat exchangers, which can be viewed in Appendix 7.3.3.

To calculate the cost of the equipment, Gavin Towler's cost values were used for each equipment. The utilities for cooling water and electricity were calculated using the yearly operating costs of 0.354\$/GJ and 16.8\$/GJ, respectively. The operating profit of liquefied methane was calculated using the price of 480\$/1000kg. All calculated costs and profits were in 2010 dollars, which were then converted to 2020 dollars. Details of all cost and profit calculations can be found in Appendix 7.1, and tabulated values in 2010 dollars for the equipment capital cost and OPEX can be found in Appendix 7.3.

Finally, to compare the Claude process to the Linde process, the split fraction to the expander was set to zero. To investigate whether there was an optimal split fraction to the expander, a sensitivity analysis was run by varying the split fraction of the expander to calculate the OPEX.

3.0 – Physical Properties

Table 1: Physical properties of the chemical components in the LNG process

Property Name	Units	CH ₄	C ₂ H ₆	C ₃ H ₈	C ₄ H _{10_1}	N ₂
Molecular Weight		16.0430	30.0700	44.0970	58.1240	28.0130
Critical Temperature	C	-82.5500	32.2500	96.6500	152.0500	-146.9500
Critical Pressure	atm	45.4000	48.2000	41.9000	37.5000	33.5000
Critical Volume	cm ³ /mol	99.2094	148.1700	203.4945	254.9212	89.6408
Compressibility Factor		0.2880	0.2850	0.2810	0.2740	0.2900
Accentric Factor		0.0080	0.0980	0.1520	0.1930	0.0400
Normal Boiling Point	C	-161.4500	-88.6500	-42.0500	-0.4500	-195.7500
Freezing Point	C	-182.4500	-183.2500	-187.6500	-138.3500	-209.8500
Liquid Molar Volume	cm ³ /mol	37.7482	55.0614	75.7832	96.5753	34.7065
Enthalpy of Formation	kcal/mol	-17.8900	-20.2400	-24.8200	-30.1500	0.0000
Gibbs Energy of Formation	kcal/mol	-12.1500	-7.8700	-5.6100	-4.1000	0.0000
Heat of Vaporization	kcal/mol	1.9550	3.5150	4.4870	5.3520	1.3330
Heat Capacity	cal/(mol C)					
A1		4.5980	1.2920	-1.0090	2.2660	7.4400
A2 x 1.0E3		12.4500	42.5400	73.1500	79.1300	-3.2400
A3 x 1.0E6		2.8600	-16.5700	-37.8900	-26.4700	6.4000
A4 x 1.0E9		-2.7030	2.0810	7.6780	-0.6740	-2.7900
A5 x 1.0E12		0.0000	0.0000	0.0000	0.0000	0.0000
A6 x 1.0E15		0.0000	0.0000	0.0000	0.0000	0.0000
Low Temperature Limit	C	26.8500	26.8500	26.8500	26.8500	26.8500
Upper Temperature Limit	C	1367.8500	1900.9501	1371.8500	1144.8500	936.0500
Liquid Vapor Pressure	Pa					
A1		43.3183	54.9873	63.5869	66.1499	46.1539
A2		-1380.5780	-2636.6211	-3550.1941	-4301.3838	-984.2006
A3		0.0000	0.0000	0.0000	0.0000	0.0000
A4		0.0116	0.0085	0.0082	0.0065	0.0232
A5		-4.3964	-5.8931	-7.0923	-7.2476	-5.4530
A6		0.0000	0.0000	0.0000	0.0000	0.0000
A7		6.0000	6.0000	6.0000	6.0000	6.0000
Low Temperature Limit	C	-182.4500	-122.6213	-83.6707	-48.8291	-209.8500
Upper Temperature Limit	C	-82.5500	32.2500	96.6500	152.0500	-146.9500

4.0 – Components

For the liquefaction of natural gas, the following components were involved:

CH₄ – Methane

C₂H₆ – Ethane

C₃H₈ – Propane

C₄H₁₀ – n-Butane

N₂ – Nitrogen

5.0 – Equipment

- *Three Stage Compressor (K-100A,B,C) Cooler I and II (E-100A, E-100B)*

Three-stage compressor with inter-stage cooling, the system has an inlet stream (S1) at 35°C and 1 bar pressure, and an output stream (S2) at 142°C and 30 bar. It requires 546 kW of power total. Each cooler reduces the outlet temperature of the previous compressor to 40°C and results in a total duty of 85934 cal/sec.

- *Pre-Cooler (E-101)*
This equipment cools down stream S2 from 142°C to 40°C with a duty of 47312 cal/sec.
- *Heat Exchanger I (E-102)*
This equipment exchanges heat between the hot and cold stream. It cools the hot stream from 40°C to 13°C, while the cold stream is heated up from 3°C to 38°C.
- *Heat Exchanger II (E-103)*
This equipment exchanges heat between the hot and cold stream. It cools the hot stream from 13°C to -94°C, while the cold stream is heated up from -145°C to 3°C.
- *Expander (EX-100)*
This equipment is responsible to reduce the pressure of split stream S5 from 30 bar down to 1 bar. As a result, the temperature also decreases from 30°C to -127°C. It requires 72kW of power.
- *Flash Tank (D-100)*
This equipment is responsible for separating the liquid product from any vapor remaining in the stream.

6.0 – Discussion and Analysis

As explained in the System and Process Section, the production of LNG using the Claude process was created in Aspen Plus V8.4. After all the specification previously explained, the simulation converged, however, the purge stream fraction is incredibly high. The resulting loss of methane is 8.8% of what enters the flash tank, D-100.

From the entire system, we recover 298 kg/hr of LNG, or about 2.4 KMTA. This product rate produces an annual revenue in 2020 dollars of \$1.6MM per operating year. As for the CAPEX, we calculate a cost of \$16.4MM in 2020 dollars. The OPEX is calculated as \$427M per operating year in 2020 dollars. This gives an annual operating profit of \$1.2MM. The costliest equipment is the three-stage compressor, which has an annualized cost of about \$4.8MM per compressor. If the power needed in those can be reduced, the CAPEX value will be decreased immensely. This will also dramatically decrease the utility cost.

Using the Linde process, we only achieved a product rate of 12.6 kg/hr of LNG. This is extremely miniscule compared to the Claude process, resulting in a 95% loss of production. The annual revenue only comes to \$68M, which is 4% of the revenue from the Claude process. This value results in a negative operating profit, which is not feasible.

As for the sensitivity analysis for the optimal value of the flow to the expander, one could not be found. As seen in Appendix 7.3.5, the minimum was achieved at around 39%. However, the inconsistencies of the sensitivity analysis made the results unreliable. Therefore, we kept the split to the expander at 35%.

To minimize the loss of the methane in the purge fraction, the only feasible solution would be to introduce the use of refrigeration, such as liquid nitrogen, into the system to use as the cold streams in the heat exchangers. Although this may cost more up front, the refrigeration can be recycled because it never makes contact with the propane. Also, this will reduce the vapor fraction going into the flash tank, therefore producing more liquid. In turn, this will yield a higher rate of LNG production.

7.0 – Appendix

7.1 – Appendix I: Formulas

Equipment Cost [1]: $Purchased\ Cost = a + bS^n$

$Installed\ Cost = (Purchased\ Cost)(Installation\ Factor)$

$$Annualized\ Cost = \frac{4}{3} (Installed\ Cost)$$

Example: Compressor K-100A (In \$2010)

$$Production\ Cost = \$580,000 + \$20,000(181.50)^{0.6} = \$1,033,269$$

$$Installed\ Cost = 2.5(\$761,287) = \$2,583,172$$

$$Annualized\ Cost = \frac{4}{3} (\$1,903,218) = \$3,444,230$$

Conversion to \$2020 from \$2010: $\$(2020) = \$(2010) \times \frac{index(2020)}{index(2010)}$

Example: Annualized Cost of K-100A

$$\$(2020) = \$3,444,230 \times \frac{2100}{1500} = \$4,821,222$$

Utility Cost: $Price\ of\ Utility = (Utility\ Cost)(Utility\ Usage)$

$Cost\ of\ Utility = (Price\ of\ Utility)(Operating\ Hours)$

Example: Electricity for K-100A

$$Price\ of\ Utility = \left(\frac{\$16.80}{GJ} \right) \left(\frac{1\ GJ}{277.78\ kW * hr} \right) (181.50\ kW) = \frac{\$10.98}{hr}$$

$$Cost\ of\ Utility = \left(\frac{\$10.98}{hr} \right) \left(\frac{8000\ hr}{yr} \right) = \$87,816/yr$$

Annual Revenue \$(2010):

$$298 \frac{kg}{hr} \left(\frac{\$480}{1000kg} \right) \left(\frac{8000\ hr}{op.\ year} \right) = \$1,144,370\ per\ operating\ year$$

Cooler Surface Area Calculation:

$$Q = F_t \cdot A \cdot U \cdot \Delta T_{ln}$$

Where Q = duty, F_t = Correction Factor, A = Surface Area, U = heat transfer coefficient, and

$$\Delta T_{ln} = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)}$$

Pressure Vessel Calculation:

$$\tau = \frac{V}{F}$$

Where τ = residence time (min), $V = 1/2 V_{\text{vessel}}$ and F = inlet flowrate (L/min)

$$\text{Diameter of the vessel} = D_i = \left(\frac{2V_{\text{vessel}}}{\pi}\right)^{1/3}$$

$$\text{Height of the vessel} = H = 1.15L = 1.15 \cdot 2D_i$$

$$\text{Thickness of the vessel} = t_w = \frac{P_i D_i}{2SE - 1.2P_i}$$

Where P_i = Allowable pressure (N/m²), S = vessel stress (ksi), E = welding efficiency

$$\text{Mass of vessel} = M = \pi \cdot D \cdot H \cdot t_w \cdot \rho_{\text{steel}}$$

Example: D-100 Size Calculation (for costing)

$$V_{\text{vessel}} = 2V = 2\tau \cdot F = 2(5 \text{ min}) \left(14293 \frac{\text{L}}{\text{min}}\right) \left(\frac{1 \text{ m}^3}{1000 \text{ L}}\right) = 142.9 \text{ m}^3$$

$$D_i = \left[\frac{2(142.9 \text{ m}^3)}{\pi}\right]^{1/3} = 4.5 \text{ m}$$

$$H = 1.15 \cdot 2(4.5) = 10.35 \text{ m}$$

$$t_w = \frac{110000 \frac{\text{N}}{\text{m}^2} (4.5 \text{ m})}{2(12.9 \text{ ksi}) \left(\frac{6.894 \frac{\text{N}}{\text{mm}^2}}{\text{ksi}}\right) \left(\frac{1000 \text{ mm}}{\text{m}}\right)^2 - 1.2 \left(11000 \frac{\text{N}}{\text{m}^2}\right)} = 2.78 \text{ mm}$$

$$M = \pi(4.5 \text{ m})(10.35 \text{ m})(0.00278 \text{ m}) \left(8000 \frac{\text{kg}}{\text{m}^3}\right) = 3254 \text{ kg}$$

7.2 – Appendix II: Process Flow Diagram (PFD)

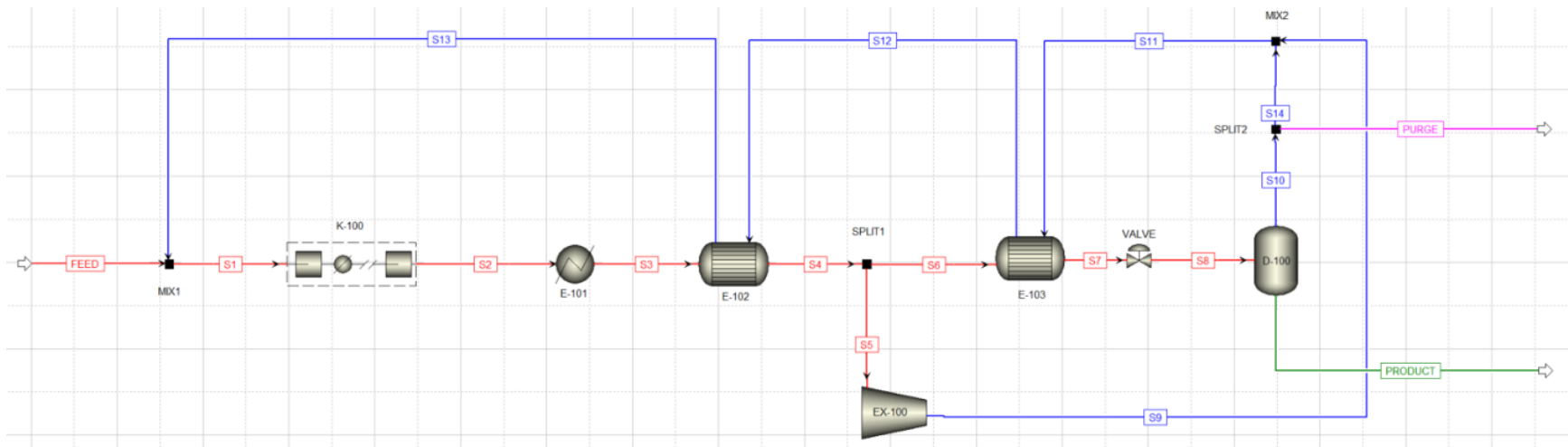


Figure 1: PFD for LNG

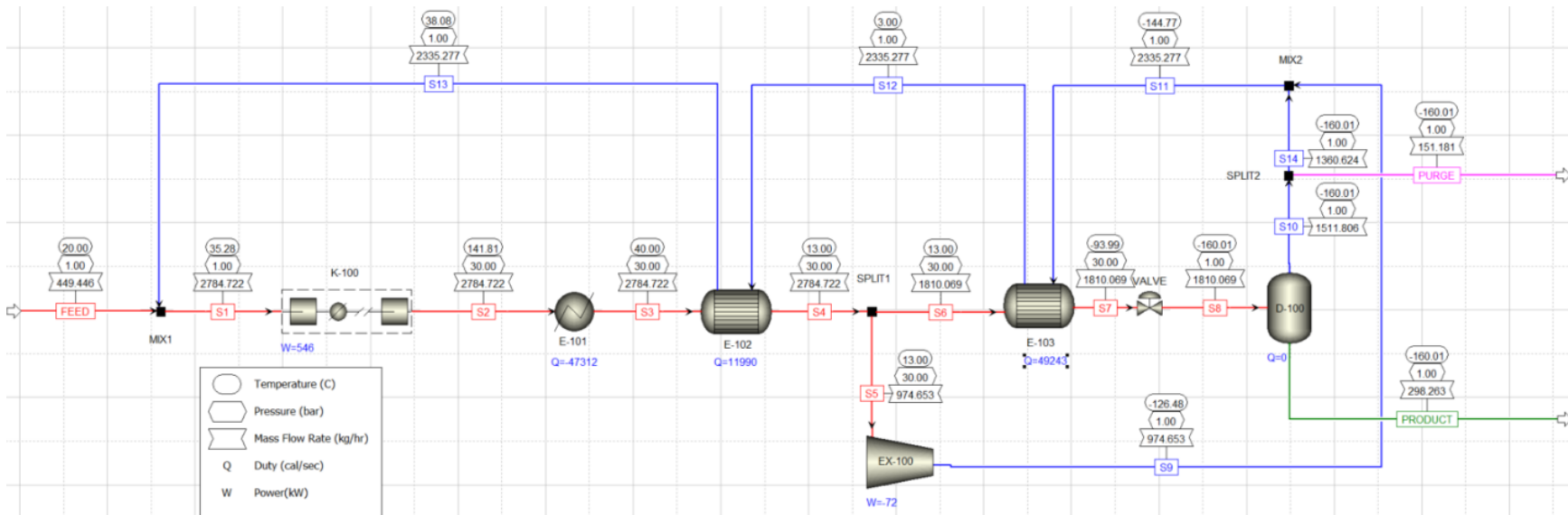


Figure 2: PFD for LNG with results

7.3 – Appendix III: Results

7.3.1 Three-Stage Compressor

Summary	Balance	Profile	Coolers	Stage Curves
Outlet pressure:	30	bar		
Total work:	546.399	kW		
Total cooling duty:	-86934.7	cal/sec		
Net work required:	546.399	kW		
Net cooling duty:	-86934.7	cal/sec		

Figure 3: Summary of results of the three-stage compressor

Summary	Balance	Profile	Coolers	Stage Curves	Wheel Curves	Utilities	 Status
	Total	Units	In	Out	Rel. diff		
▶	Mole-flow	kmol/hr	168.675	168.675	0		
▶	Mass-flow	kg/hr	2784.73	2784.73	0		
▶	Enthalpy	cal/sec	-832618	-789047	-0.0523294		

Figure 4: Mass and mole balance for the three-stage compressor

Summary		Balance	Profile	Coolers	Stage Curves	Wheel Curves	Utilities	Status	
	Stage	Temperature	Pressure	Pressure ratio	Indicated power	Brake horsepower	Head developed	Volumetric flow	Efficiency used
		C	bar		kW	kW	m-kgf/kg	l/min	
▶	1	134.711	3.10723	3.10723	181.545	181.545	20342.4	71940.5	0.85
▶	2	140.597	9.65489	3.10723	183.476	183.476	20558.7	23411.3	0.85
▶	3	141.806	30	3.10723	181.377	181.377	20323.6	7437.95	0.85

Figure 5: Profile results for the three-stage compressor

Summary		Balance	Profile	Coolers	Stage Curves	Wheel Curves
	Stage	Temperature	Pressure	Duty	Vapor fraction	
		C	bar	cal/sec		
▶	1	40	2.41776	-7677.1	1	
▶	2	40	8.96542	-12794.8	1	
▶	3	142.508	30	0	1	

Figure 6: Result for the inter-stage cooling for the three-stage compressor

Compressor utility usage					
Utility: ELECTRIC	Stage	Duty cal/sec	Usage kW	Cost \$/hr	CO2 emission rate kg/hr
	1	43361.3	181.545	10.9799	62.9786
	2	43822.5	183.476	11.0966	63.6484
	3	43321.2	181.377	10.9697	62.9203

Cooler utility usage					
Utility: COOLWAT	Stage	Duty cal/sec	Usage kg/hr	Cost \$/hr	CO2 emission rate
	1	-41827.4	30199.8	0.223177	
	2	-45107.3	32567.8	0.240677	
	3	0	0	0	

Figure 7: Results for the utility used in the three-stage compressor

7.3.2– Heat and Weight Balance

Material	Heat	Load	Work	Vol.% Curves	Wt. % Curves	Petroleum	Polymers	Solids
Display: Streams Format: GEN_E Stream Table Copy All								
		FEED	PRODUCT	PURGE				
▶	Temperature C	20	-160	-160				
▶	Pressure bar	1	1	1				
▶	Vapor Frac	1	0	1				
▶	Mole Flow kmol/hr	25	15.596	9.404				
▶	Mass Flow kg/hr	449.445	298.264	151.181				
▶	Volume Flow l/min	10125.6	10.443	1428.32				
▶	Enthalpy MMBtu/hr	-1.811	-1.391	-0.719				
▶	Mole Flow kmol/hr							
▶	METHANE	22.57	13.191	9.379				
▶	ETHANE	1.583	1.581	0.002				
▶	PROPANE	0.623	0.622	trace				
▶	BUTANE	0.2	0.2	trace				
▶	NITROGEN	0.025	0.001	0.024				

Figure 8: Heat and mass balance for the feed, product and purge streams

7.3.3– TEMA Sheet Analysis

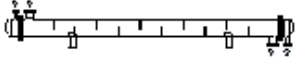
6	Size	307 - 6000	mm	Type	BEM	Hor	Connected in	1 parallel	3 series			
7	Surf/unit(eff.)	104.1	m²	Shells/unit	3		Surf/shell (eff.)	34.7	m²			
8	PERFORMANCE OF ONE UNIT											
9	Fluid allocation				Shell Side		Tube Side					
10	Fluid name				HEX1.S12		HEX1.S3					
11	Fluid quantity, Total	kg/s				0.5631		0.688				
12	Vapor (In/Out)	kg/s				0.5631	0.5631	0.688	0.688			
13	Liquid	kg/s				0	0	0	0			
14	Noncondensable	kg/s				0	0	0	0			
15												
16	Temperature (In/Out)	°C				3	38.9	40	12.95			
17	Dew / Bubble point	°C										
18	Density Vapor/Liquid	kg/m³				0.71 /	0.24 /	20.24 /	22.57 /			
19	Viscosity	mPa s				0.0105 /	0.0116 /	0.0121 /	0.0113 /			
20	Molecular wt, Vap				16.29		16.29		16.57			
21	Molecular wt, NC								16.57			
22	Specific heat	kJ/(kg K)				2.167 /	2.246 /	2.41 /	2.396 /			
23	Thermal conductivity	W/(m K)				0.0309 /	0.0359 /	0.0357 /	0.0319 /			
24	Latent heat	kJ/kg										
25	Pressure (abs)	bar				1	0.37926	30	29.89692			
26	Velocity	m/s				116.93		2.01				
27	Pressure drop, allow./calc.	bar				0.68948	0.62074	0.68948	0.10308			
28	Fouling resistance (min)	m² K/W				0.00018		0.00018	0.00023 Ao based			
29	Heat exchanged	44.6	kW				MTD corrected		3.93 °C			
30	Transfer rate, Service	109	Dirty	110.2	Clean	115.4	W/(m² K)					
31	CONSTRUCTION OF ONE SHELL							Sketch				
32			Shell Side		Tube Side							
33	Design/vac/test pressure:g	bar	3/	/	33/	/						
34	Design temperature	°C	75		75							
35	Number passes per shell		1		1							
36	Corrosion allowance	mm	3.18		3.18							
37	Connections	In	mm	1	152.4/	-				1	76.2/	-
38	Size/rating	Out	1	203.2/	-	1				50.8/	-	
39	Nominal	Intermediate	1	152.4/	-	1	50.8/	-				
40	Tube No.	98	OD	19.05	TksAvg	2.11	mm	Length	6000			
41	Tube type	Plain	#/m	Material	Carbon Steel	Tube pattern		30				
42	Shell	Carbon Steel	ID	307.09	OD	323.85	mm	Shell cover	-			
43	Channel or bonnet	Carbon Steel				Channel cover		-				
44	Tubesheet-stationary	Carbon Steel				Tubesheet-floating		-				
45	Floating head cover	-				Impingement protection		None				
46	Baffle-cross	Carbon Steel	Type	Single segmental	Cut(%d)	43.28	H	Spacing: c/c	500			
47	Baffle-long	-	Seal type				Inlet	707.48	mm			
48	Supports-tube	U-bend	0		Type							
49	Bypass seal					Tube-tubesheet joint		Exp. 2 grv				
50	Expansion joint	-	Type				None					
51	RhoV2-Inlet nozzle	1283	Bundle entrance		234	Bundle exit		698	kg/(m s²)			
52	Gaskets - Shell side	-	Tube Side				Flat Metal Jacket Fibe					
53	Floating head	-										
54	Code requirements	ASME Code Sec VIII Div 1					TEMA class	R - refinerv service				
55	Weight/Shell	1220.4	Filled with water		1615.2	Bundle	615.3	kg				

Figure 9: TEMA analysis for E-102.

6	Size	387 - 6000	mm	Type	BEM	Hor	Connected in	1 parallel	5 series
7	Surf/unit(eff.)	321.6	m²	Shells/unit	5		Surf/shell (eff.)	64.3	m²
8	PERFORMANCE OF ONE UNIT								
9	Fluid allocation			Shell Side			Tube Side		
10	Fluid name			HEX2.S11			HEX2.S6		
11	Fluid quantity, Total			0.5631			0.4472		
12	Vapor (In/Out)			0.5489			0.4472		
13	Liquid			0.0141			0		
14	Noncondensable			0			0		
15									
16	Temperature (In/Out)			-151.48			13		
17	Dew / Bubble point			-105.64			-62.84		
18	Density Vapor/Liquid			1.63 / 589.82			0.25 / 22.65		
19	Viscosity			0.0048 / 0.358			0.0104 / 0.0113		
20	Molecular wt, Vap			16.08			16.29		
21	Molecular wt, NC								
22	Specific heat			2.112 / 2.421			2.165 / 2.397		
23	Thermal conductivity			0.0134 / 0.1975			0.0307 / 0.0319		
24	Latent heat			513.2			457.3		
25	Pressure (abs)			1			0.35648		
26	Velocity			90.48			0.63		
27	Pressure drop, allow./calc.			0.68948			0.64352		
28	Fouling resistance (min)			0.00018			0.00018		
29	Heat exchanged			189.8			MTD corrected		
30	Transfer rate, Service			60.1			Clean 63.2		
31	CONSTRUCTION OF ONE SHELL								
32				Shell Side			Tube Side		
33	Design/vac/test pressure:g			3/			33/		
34	Design temperature			50			50		
35	Number passes per shell			1			1		
36	Corrosion allowance			3.18			3.18		
37	Connections			In			1		
38	Size/rating			Out			1		
39	Nominal			Intermediate			1		
40	Tube No.			182			OD 19.05		
41	Tube type			Plain			#/m		
42	Shell			Carbon Steel			ID 387.35		
43	Channel or bonnet			Carbon Steel			OD 406.4		
44	Tubesheet-stationary			Carbon Steel			-		
45	Floating head cover			-			Impingement protection		
46	Baffle-cross			Carbon Steel			Type		
47	Baffle-long			-			Seal type		
48	Supports-tube			U-bend			0		
49	Bypass seal			-			Type		
50	Expansion joint			-			Type		
51	RhoV2-Inlet nozzle			544			Bundle entrance		
52	Gaskets - Shell side			-			Tube Side		
53	Floating head			-			Bundle exit		
54	Code requirements			ASME Code Sec VIII Div 1			TEMA class		
55	Weight/Shell			1940.4			Filled with water		

Figure 10: TEMA analysis for E-103.

7.3.4 – CAPEX and OPEX Book

 Chemical Engineering Department UNIVERSITY OF ILLINOIS at CHICAGO				Assignment or Project Description:								
				Assignment Number # 6: Liquefaction of Natural Gas with the Claude Process. Main goal of the assignment is the detailed design of Heat Exchangers as well as the estimation of the total cost of production.								
Student Names		James Fell, William Rawson, Mariana St. Peter, Jasmine Villegas					Capital Cost Basis Year		2010			
Plant Location		Chicago, IL					Units		Metric			
Date of Completion		Wednesday, December 4, 2019					On Stream		8,000			
Equipment Class	ASPEN Block	Description	Material of Construction	Operating Pressure (bar)	Diameter (meter) or Volume (cum)	# Trays or Height (meter)	Surface Area (sqmeter)	Power (kW)	Volumetric Flow (cum/hr) or Mass Flow (kg/hr)	Purchased Cost	Installed Cost	Annualized Cost
Distillation Columns												
Pressure Vessels and Drums												
Vertical Vessel	D-100	Seprates gas and liquid products	Carbon Steel	1.10	4.50	10.35				\$ 44,010.30	\$ 176,041.19	\$ 234,721.58
Heat Transfer Equipment												
U-Tube S&T	E-100A	Cools fluid after K-100A	Carbon Steel				8.31			\$ 28,685.45	\$ 100,399.09	\$ 133,865.45
U-Tube S&T	E-100B	Cools fluid after K-100B	Carbon Steel				8.59			\$ 28,713.16	\$ 100,496.05	\$ 133,994.74
U-Tube S&T	E-101	Cools Stream S2	Carbon Steel				8.93			\$ 28,747.16	\$ 100,615.07	\$ 134,153.43
U-Tube S&T	E-102	Cools Stream S3	Carbon Steel				104.10			\$ 42,234.25	\$ 147,819.88	\$ 197,093.17
U-Tube S&T	E-103	Cools Stream S6	Carbon Steel				321.60			\$ 83,102.72	\$ 290,859.51	\$ 387,812.68
Pumps and Drivers												
Turbines												
Condensing Steam Turbine	EX-100	Expands Stream S5	Carbon Steel					72.00		\$ 32,962.69	\$ 131,850.76	\$ 175,801.02
Compressors												
Centrifugal	K-100A	Compresses Stream S1	Carbon Steel					181.50		\$ 1,033,268.95	\$ 2,583,172.38	\$ 3,444,229.85
Centrifugal	K-100B	Compresses Stream S1	Carbon Steel					183.50		\$ 1,036,259.20	\$ 2,590,648.01	\$ 3,454,197.34
Centrifugal	K-100C	Compresses Stream S1	Carbon Steel					181.40		\$ 1,033,119.10	\$ 2,582,797.74	\$ 3,443,730.32
Reactors												
Misc.												
TOTAL EQUIPMENT COST										\$ 3,391,102.98	\$ 8,804,699.68	\$ 11,739,599.57
Utility Class	ASPEN Block	Description				Q Value (GJ/hr)	Heating Value (\$/GJ)	Usage (units/hr)	Annual Plant Usage	Price of Utility (\$/unit)	Unit of Price	Cost of Utility
Steam												
Hot Oil or Fuel Gas												
Cooling Water												
Cooling water	E-100A	Cooling water used in E-100A				0.63	0.35	0.63	5043.54	\$ 0.22	\$/GJ	\$ 1,785.41
Cooling water	E-100B	Cooling water used in E-100B				0.68	0.35	0.68	5439.02	\$ 0.24	\$/GJ	\$ 1,925.41
Cooling water	E-101	Cooling water used in E-101				0.71	0.35	0.71	5704.90	\$ 0.25	\$/GJ	\$ 2,019.54
Electricity												
Electric	K-100A	Electricity used in K-100A				0.65	16.80	0.65	5228.49	\$ 10.98	\$/GJ	\$ 87,838.60
Electric	K-100B	Electricity used in K-100B				0.66	16.80	0.66	5284.10	\$ 11.10	\$/GJ	\$ 88,772.81
Electric	K-100C	Electricity used in K-100C				0.65	16.80	0.65	5223.66	\$ 10.97	\$/GJ	\$ 87,757.42
Please Select Utility Type						0.26	16.80	0.26	2074.03	\$ 4.36	\$/GJ	\$ 34,843.78
TOTAL UTILITY COST												\$ 304,942.97

Figure 11: Cost book for CAPEX and OPEX calculations in 2010 dollars.

7.3.5 – Sensitivity Analysis for Optimal Flow Fraction

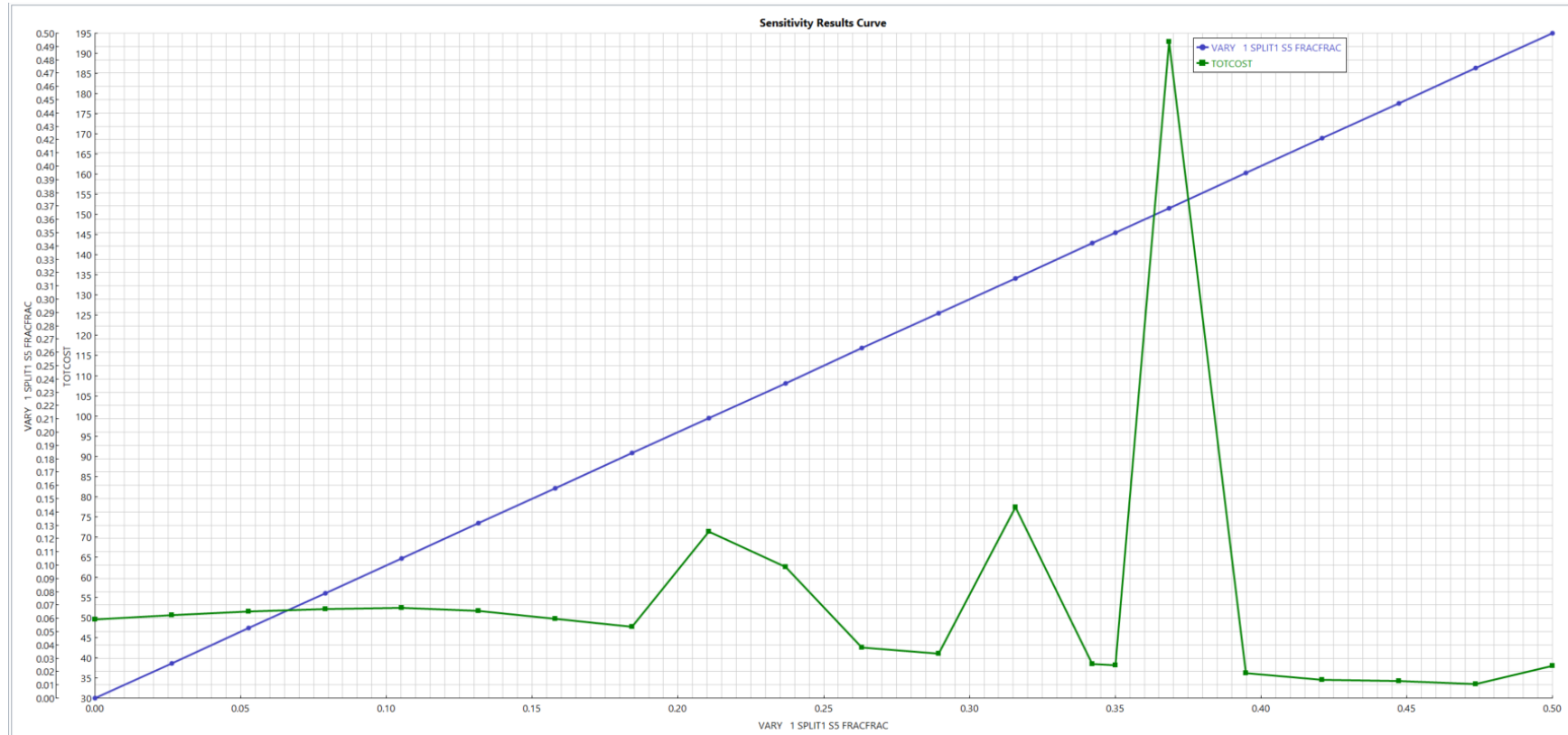


Figure 12: Sensitivity analysis for the total utility cost by varying the split fraction to EX-100

8.0 – References

[1] Towler, Gavin and Sinnott, Ray. Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design. Prentice Hall, 2nd Edition. 2012.